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ESPR preparedness in cutting tool development: Insights from battery and steel regulation

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Preface

This bachelor's thesis is the concluding part of our studies in the bachelor's Programme in Industrial Engineering and Management at the University of Gävle, comprising 15 credits. The research has been carried out in collaboration with Seco Tools, and it explores the intersection of product development, sustainability, and upcoming European regulations. The work has been highly educational, challenging, and has provided us with a profound understanding of the complexities surrounding life cycle assessments and circular economy within the manufacturing industry.

We would like to extend our sincere gratitude to Seco Tools for the opportunity to conduct this thesis with you. A special thanks goes to interviewees from the product management, research and development, and sustainability teams, who generously shared their time, expertise, and valuable insights, which were crucial for the completion of this study.

We also want to thank our academic supervisor, Lars Löfqvist, and our examiner, Stefan Eriksson, at the University of Gävle for their continuous support, constructive feedback, and guidance throughout the research process.

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Abstract

The European manufacturing industry is undergoing a transition towards more sustainable and circular production, driven in part by the Ecodesign for Sustainable Products Regulation (ESPR). For cutting tool manufacturers, however, the specific requirements that will apply to their products remain uncertain because product-specific delegated acts have not yet been established. This creates a need to anticipate how ESPR requirements may affect product development and how environmental performance can be assessed during the use phase.

The purpose of this thesis is to investigate how the upcoming ESPR may influence the development of cutting tools from a life cycle perspective and to identify measurable use phase product performance parameters that can support future product development. A qualitative case study was conducted in collaboration with Seco Tools. The study combines a review of ESPR-related regulatory frameworks and precedents from the battery and steel industries with semi-structured interviews involving five representatives from product management, research and development, sustainability, sustainability compliance, and sustainability and EHS.

The findings indicate that cutting tools currently lack finalized product specific requirements and standardized methods for assessing use phase environmental performance. Nevertheless, the study identifies four preliminary measurement areas: use phase energy efficiency, climate impact, tool lifetime and circularity related material data. The interviews further show that application-specific assessment is important because cutting tool performance depends on machining conditions, tool life, cycle time, cutting forces and material removal. Significant implementation challenges were also identified in relation to data fragmentation, supplier traceability and the connection between environmental information and individual products.

The study concludes that ESPR preparedness for cutting tools should focus on establishing reliable, application-specific performance data and integrating sustainability related information into existing product development processes. The identified measurement areas should be regarded as preliminary rather than finalized regulatory requirements and provide a basis for further validation as product specific ESPR requirements are developed.

Keywords: *ecodesign, ESPR, cutting tools, life cycle assessment, use phase, digital product passport, sustainable product development*

Table of contents

Preface i
Abstract ii
1. Introduction and background 1
1.1 Problem formulation 1
1.2 Purpose statement and research questions 1
2. Theoretical and regulatory framework 3
2.1 Ecodesign for Sustainable Products Regulation (ESPR) 3
2.1.1 The Critical Raw Materials Act 5
2.1.2 LESS 6
2.1.3 The steel decarbonization scale 6
2.1.4 Adjustment rules for quality steel 7
2.2 Life cycle assessment 7
2.3 The NPD process structure and its agile evolution 8
2.4 Eco Stage Gate: Integrating sustainability and environmental goals 9
2.5 Cutting tools theory 10
2.5.1 Types of cutting tools 10
3. Methodology 13
3.1 Contact with Seco Tools 13
3.2 Scientific approach 13
3.3 Data collection 13
3.4 Interviews 14
3.5 Research ethics 14
4. Results and practical implications 15
4.1 ESPR in the battery and steel industries 15
4.1.1 Synthesized performance metrics 15
4.1.2 ESPR In the battery industry 16
4.1.3 ESPR in iron and steel industries 17
4.2 Seco Tools and their work with sustainability 18
4.3 Interviews from the team's triangulation 21
4.3.1 Product management team 21
4.3.2 Research and development team 22
4.3.3 Sustainability team 23
4.3.4 Sustainability compliance officer, insights in DPP 25
4.3.5 Sustainability and EHS Council Execution Lead 26
5. Analysis and discussion 28
5.1 ESPR in the battery and steel industry 28

5.2 Resolving the functional unit paradox via PEF 28
5.3 Overcoming scrap scarcity with LESS adjustments 29
5.4 Digital Product Passports and data management complexities 30
5.5 Learnings from the ESPR regulation 31
5.5.1 Technological manifestations of ESPR parameters 31
5.5.2 Integrating ESPR into the product development process 33
5.5.3 Critical discussion and limitations 34
6. Conclusion 35
6.1 Answer to research question 1 35
6.2 Answer to research question 2 36
6.3 Theoretical contributions 37
6.4 Practical contributions 37
6.5 Limitations of the study 38
6.6 Final conclusion 38
References 39
Appendices: Interview questions 42

1. Introduction and background

1.1 Problem formulation

This bachelor's thesis is carried out in collaboration with Seco Tools, which has its headquarters in Fagersta, Sweden and is one of the world's leading providers of complete solutions for cutting machining, offering a comprehensive range for milling, turning, drilling bits and tooling systems (Seco Tools, n.d.). The company is part of the Sandvik Group, which has an innovative history dating back to 1862 (Seco Tools, n.d.). Seco Tools itself has over 80 years of experience in the industry, with the development of cemented carbide tools for metalworking gaining momentum during the 1950s (Seco Tools, n.d.). In today's operations, there is a strong focus on sustainability and innovation, reflected in services such as tool recycling and reconditioning to reduce environmental impact. As a global player in a changing industry, the company now faces new challenges linked to the green transition and increasingly stringent regulatory requirements.

The European manufacturing industry is currently undergoing a major transformation toward a circular economy, a development driven by new EU legislation such as the ESPR (Ecodesign for Sustainable Products Regulation) (Faraca et al., 2024). For manufacturers like Seco Tools, this means new requirements for tracing the entire supply chain, from raw material extraction to final production. A critical issue, however, is the current regulatory uncertainty, as specific delegated acts for cutting tools have not yet been established despite the ESPR's major impact on the industry. Consequently, key sustainability metrics during the product use phase, such as carbon footprint per component, carbon footprint per volume of material removed, or the total tool life remain undefined in the current legislative landscape. The challenge therefore lies in proactively adapting product development processes and sustainability frameworks without clear legal guidelines to follow.

The need for this bachelor's thesis rests on several strategic and operational factors, where proactive adaptation is crucial for Seco Tools to maintain its market leading position by developing its own models for measuring environmental performance before legislation takes effect. There is also a growing demand for transparency and traceability from both customers and authorities regarding the product's entire lifecycle and its environmental impact at every stage. By defining relevant key performance indicators for sustainable innovation, the company can steer its product development toward solutions that are both technically advanced and optimized for a circular economy. Finally, the work contributes to necessary standardization by filling the knowledge gap between overarching EU directives and practical application in cutting machining, which is a prerequisite for future sustainable production.

1.2 Purpose statement and research questions

The purpose of this thesis is to investigate how the upcoming ESPR impacts the development of cutting tools from a life cycle perspective. By analyzing the established regulatory frameworks and learnings from the battery and steel industries as precedents, the study aims to identify measurable use phase performance parameters that can be anticipated and

practically applied within Seco Tools product development process. This thesis will work on the research questions below.

1. What requirements do ESPR and associated standards impose on cutting tools from a life cycle and Ecodesign perspective, based on learnings from the battery and steel industries?
2. What measurable use phase product performance parameters can be applied in the product development of cutting tools?

2. Theoretical and regulatory framework

2.1 Ecodesign for Sustainable Products Regulation (ESPR)

Regulation (EU) 2024/1781 of the European Parliament and of the Council of 13 June 2024 establishing a framework for the setting of Ecodesign requirements for sustainable products (ESPR) constitutes the cornerstone of EU market legislation. By mandating specific Ecodesign criteria, this framework guarantees that all products sold within the internal market actively support the Union's strategic environmental and climate targets. The European Green Deal highlights the urgent need to transition away from today's linear "take-make-waste" patterns. Current systems are built by a resource-intensive cycle where raw materials are extracted and sold, only to be converted into waste within a short timeframe. The ESPR addresses these systemic flaws by moving beyond traditional energy efficiency during the usage phase. Instead, it mandates that circularity and resource efficiency be explicitly integrated throughout the entire economic value chain.

The ESPR grants the European Commission the authority to implement delegated acts for various product categories. These standards are classified into two main pillars, namely performance requirements and information requirements. Performance requirements are defined as either quantitative or non-quantitative mandates imposed on a product to ensure a specific level of achievement. These requirements are always linked to a particular product parameter to drive measurable environmental improvements. In practice, this may entail establishing minimum thresholds for energy efficiency and strict maximum limits for the carbon footprint per functional unit. It may also include binding criteria regarding the minimum share of recycled input required in the product. Under the regulation, information requirements establish a formal obligation to accompany the product with specific documentation. This includes detailed data on the product's environmental performance, alongside explicit instructions for its installation, maintenance, repair, and eventual dismantling and recycling.

One of the most pivotal aspects of the ESPR is the formal prohibition against premature obsolescence. This means companies can no longer use built-in features, software updates, or deliberate neglect to make a product breakdown faster than it naturally should. The framework dictates that Member States must not obstruct, restrict, or forbid the commercialization of products, provided these items satisfy the specific performance standards defined in the applicable delegated acts.

To formulate and detail these ecodesign requirements, the Commission relies on a predefined list of product parameters. These parameters provide the technical baseline necessary to drive a product's environmental sustainability. The regulation establishes the following parameters, which are highly relevant to the manufacturing sector (Table 1).

Table 1:Product parameters for sustainability

Product parameter	**Description and analytical significance**

Durability and reliability	The parameters are defined by the product's guaranteed lifetime and technical durability, including the mean time between failures. Furthermore, they mandate data on the actual operational environment, structural resilience under load, and overall resistance to wear and tear.
Repairability and maintenance	These parameters are expressed in terms of the characteristics, availability, delivery times, and affordability of spare parts. They also encompass design modularity, compatibility with commonly available tools and components, as well as access to comprehensive repair and maintenance instructions.
Reusability and upgradability	These parameters evaluate the potential for upgrading, reuse, remanufacturing, and refurbishment. Key metrics include the total number of materials and components involved, the reliance on standard parts, the use of established coding standards for material traceability, and the ability to perform nondestructive dismantling.
Presence of substances of concern	These criteria require strict avoidance of design choices or technical solutions that compromise reusability and recyclability. This includes the explicit tracking of harmful substances, particularly those that impede material recovery or amplify environmental footprint during manufacturing.
Resource use	This parameter addresses the consumption of energy, water, and other critical resources throughout one or more life cycle phases. This includes also the direct influence of physical factors on the product's overall efficiency.
Recycled content	This covers the amount of recycled material used and how easily materials can be recovered, especially strategic and critical raw materials.
Environmental footprint	Quantifying the product's projected waste and emissions in order to minimize its overall carbon and environmental footprint across the entire life cycle.

2.1.1 The Critical Raw Materials Act

While the ESPR establishes general Ecodesign and environmental performance mandates, Regulation (EU) 2024/1252 of the European Parliament and of the Council of 11 April 2024 establishing a framework for ensuring a secure and sustainable supply of critical raw materials (CRMA) safeguards the underlying material requirements. It addresses the fundamental supply of materials that enable green and digital transitions. The functioning of the internal market relies heavily on the availability of raw materials. The regulation classifies a distinct group of non-energy and non-agricultural raw materials as critical. This classification is based on their profound economic impact coupled with a high vulnerability to supply disruptions. Supply risks often stem from a highly concentrated global supply chain dominated by a few third countries. For advanced industrial applications like defense and aerospace, the demand for these materials is anticipated to rise at an exponential rate

Although there is considerable overlap, the CRMA establishes a clear distinction between strategic and critical raw materials. Annex I details strategic materials such as cobalt, copper, gallium, lithium, nickel, and platinum group metals. In contrast, the broader critical raw materials list adds elements like tungsten, titanium, tantalum, and scandium. From an industrial perspective, tungsten and cobalt are highly relevant for the fabrication of hard metals and precision cutting tools.

Product Environmental Footprint (PEF) and the Delta Approach

To further harmonize the calculation of environmental performance and combat "greenwashing," the European Commission has developed the Product Environmental Footprint (PEF) method, as set out in Commission Recommendation (EU) 2021/2279 of 15 December 2021 on the use of the Environmental Footprint methods. The PEF framework builds upon ISO 14040/14044 but mandates much stricter and more detailed modeling rules. These measures are designed to guarantee that environmental claims for products on the European market are truly comparable. The development of Product Environmental Footprint Category Rules (PEFCRs) provides the framework for precisely modeling the life cycle of a given product category. Each PEFCR sets out system boundaries, determines key environmental impact categories, and specifies the necessary data quality and types.

In the context of the PEF framework, the requirements for modeling the use phase are very precise, carrying weight for industrial equipment and tools. The methodology categorizes use-phase processes into two separate types. Processes that are directly or indirectly determined or influenced by the product’s design, characteristics, or instructions for use are classified as product-dependent processes. Because these processes depend on the product's characteristics, they help differentiate one product from another, for instance the electrical consumption of equipment during operation. It is strictly mandatory to include these processes within the system boundaries. Processes that are completely unrelated to how a product is designed are classified as product-independent processes. The environmental impact of these processes remains constant regardless of the specific product chosen, such as the glass used for drinking wine. These processes shall be excluded from the system boundary, but they can still be documented on a qualitative basis.

The PEF method provides two approaches for allocation in the use phase. The first is the Main function approach, where all environmental impacts during use are attributed to the product. The second is the Delta approach, which is applied when a product causes an overconsumption or underconsumption of related resources, such as a substandard tool generating excessive material waste during machining. This method relies on setting a reference consumption point, allowing the analysis to allocate the resulting excess environmental footprint to the product that created the resource inefficiency.

2.1.2 LESS

The Low Emission Steel Standard (LESS) and the concept of the "Steel Decarbonization Scale" provide an advanced framework for managing the complexities of industrial environmental reporting. The LESS framework addresses the issue of carbon intensity, raw material origins, circularity, and the technical differences between various product grades **(**Responsible Steel & Low Emission Steel Standard (LESS), 2025a**)**.

One consistent theme within the LESS framework highly relevant to all metalworking sectors is the scientific realization that the supply of recycled materials is fundamentally limited. To create performance metrics that do not distort the market, the lifespan of the products must be considered. The average lifespan of steel utilized in infrastructure and industrial products is estimated at roughly 35 years, which creates a fundamental limit on the availability of scrap material. Mathematically speaking, the maximum potential supply of consumer scrap currently available is strictly limited to the total quantity of steel produced roughly 35 years ago. Since global steel production was 0.78 billion tonnes in 1989 while demand reached 1.9 billion tonnes in 2024, the theoretical maximum recycled content for steel produced in 2024 is locked at 41% (ResponsibleSteel & LESS, 2025a). Even with an outstanding global steel recycling rate of 85%, the currently available scrap metal can only meet 32% of the world's present demand for new steel production (ResponsibleSteel & LESS, 2025a). In addition to this, unavoidable material losses occur during the use phase due to erosion and wear at an estimated <1 %, alongside losses of approximately 5 % during the conversion process at the melting stage (ResponsibleSteel & LESS, 2025a).

These measurable physical limitations result in a critical analytical realization for developing performance metrics. It becomes clear that a framework or ecolabel focusing strictly on the absolute Product Carbon Footprint would generate significant market distortions. If legislation fails to account for the physical shortage of scrap metal, it will simply create a “fruitless competition" for limited recycled materials (ResponsibleSteel & LESS, 2025a).

2.1.3 The steel decarbonization scale

To resolve this issue, the International Energy Agency (IEA) introduced the "Steel Decarbonization Scale", or "Sliding Scale", which has now been embraced by key stakeholders including ResponsibleSteel, LESS, and the China Iron and Steel Association (CISA) (ResponsibleSteel & LESS, 2025b). The scale is a measurable, technology-neutral approach that adjusts a product's carbon footprint based on the amount of scrap used as input material (ResponsibleSteel & LESS, 2025b). According to the LESS regulations the scrap share represents the ratio of the used scrap volume against all metallic inputs, consisting of scrap, pig

iron, DRI, and HBI (LESS aisbl, 2025a). The classification system consists of tier's ranging from the highest class "A" where the lowest emissions are permitted down to class "E" (LESS aisbl, 2025a). The thresholds between levels are not static but act as linear equations where allowed emissions decrease as the scrap share increases (ResponsibleSteel & LESS, 2025b).

2.1.4 Adjustment rules for quality steel

A major challenge in evaluating manufacturing environmental performance is the varying inherent technical complexity of the products. Because manufacturing products with advanced technical specifications like cutting tools or special steel inevitably consumes more energy and requires a higher share of alloying elements, it fundamentally results in a larger overall carbon footprint. If the classification were based entirely on absolute emission levels, manufacturers of highly technical components would face a systematic disadvantage relative to manufacturers of basic products. The LESS rulebook effectively solves this problem by setting strict parameters for Quality Steel. Quality steel is classified as steel featuring a minimum elemental metallic alloy content of 8%, or steel meant for applications that demand high surface quality and machinability (LESS aisbl, 2025b).

To prevent the production of Quality Steel from being penalized due to the embedded energy and massive Scope 3 emissions associated with heavy alloying elements, LESS has implemented specific adjustment rules. These rules allow manufacturers to create an adjusted equivalence value for their emissions. They are allowed to subtract the Scope 3 emissions caused by necessary alloying elements, under the condition that these levels exceed the emissions from the system's designated reference plant (LESS aisbl, 2025a). Similar adjustment rules exist to account for excess input caused by material loss during the rolling of complex geometries (LESS aisbl, 2025a).

These adjustments fundamentally streamline the measurement and classification framework to highlight the process efficiency and decarbonization of the primary producer. This ensures the system does not merely measure the unavoidable material content required to achieve the product's critical functional performance during its use phase.

2.2 Life cycle assessment

Over the past several decades the Life cycle assessment has advanced dramatically, ultimately establishing itself as a foundational instrument for corporate environmental management. Tracing its origins back to the late 1960s and 1970s, a period when its application was primarily concentrating on mere energy and resource requirements. Driven by the collaborative influence of the Society of Environmental Toxicology and Chemistry and the International Organization for Standardization, the 1990s marked a pivotal era of consolidation for the framework. By establishing a unified theoretical framework and standardized terminology during this period the industry effectively eliminated the inconsistency that had formerly delayed LCA's recognition as a universally validated analytical method (Guinée et al., 2011).

The international ISO 14040 standard formally defines LCA as the systematic compilation and evaluation of a product system's inputs, outputs, and potential environmental impacts. It evaluates all material flows across the value chain, tracking the product from initial raw material

extraction through to its ultimate disposal (SS-EN ISO 14040:2006). A defining characteristic of LCA is its comparative approach, which relies heavily on a specific "functional unit". This metric defines the product's functional output and acts as a universal reference, enabling direct and objective comparisons between distinct product systems (SS-EN ISO 14040:2006).

The formal LCA procedure operates through a four-phase iterative cycle. The process commences with *the goal and scope definition*, where the project's intended use, strict system boundaries, and functional unit are officially established. The second phase known as the *Life cycle inventory analysis* centers on rigorous data collection. This critical phase is dedicated to carefully tracking the intake of energy and materials as well as the environmental outputs of waste and emissions. Moving to the third phase, that is the l*ife cycle impact assessment* which is designed to evaluate the environmental footprint. This is achieved by linking the previously gathered inventory data to specific environmental impact categories, such as global warming potential or eutrophication. In the concluding phase, that is called l*ife cycle interpretation* summarizes the results derived from both the inventory analysis and the impact assessment. This critical final step helps ensure logical conclusions and strategic recommendations that remain entirely consistent with the initial goal and scope (ISO 14044:2006).

In modern industrial operations LCA is recognized as a vital instrument. It effectively bridges the gap between the precise measurement of environmental impacts and the execution of sustainable manufacturing strategies (de Moraes et al., 2025). LCA is widely recognized for its ability to systematically diagnose environmental performance through hard data, but current research highlights that measurement alone is insufficient. Studies stress the critical need to integrate LCA with corporate solutions like Ecodesign. Integrating LCA and Ecodesign into the early phases of development allows for the strategic optimization of environmental metrics. This enables organizations to significantly minimize their water and energy consumption without sacrificing a drop in product quality or performance (de Moraes et al., 2025). Future trajectories indicate a continuous broadening of the LCA framework. It is progressively evolving into l*ife cycle sustainability assessment*, a unified framework that tackles environmental, economic, and social impacts all at once (Guinée et al., 2011).

2.3 The NPD process structure and its agile evolution

New Product Development (NPD) demands processes that are both structured and adaptable to successfully guide projects from initial concept through to market launch. A prominent framework in this domain is the Stage-Gate model, functioning as a conceptual and operational blueprint to optimize efficiency and mitigate risk throughout the product development lifecycle. Fundamentally, this model is structured around cross-functional stages, facilitating concurrent activities and the systematic collection of vital project information. Each subsequent stage necessitates a progressively larger financial commitment, thereby establishing a structured framework for incremental decision making. Following every stage is a "gate" that acts as both a quality checkpoint and a critical decision point. At these crucial checkpoints, key resource owners' step in to evaluate the project's progress against predetermined objective standards. This group makes decisive investment decisions, determining if a project should be approved, rejected, or undergo revisions before receiving further funding (Cooper, 2008).

Despite its dominance in industry, Stage-Gate is often misunderstood as a rigid, overly bureaucratic, and strictly linear process. On the contrary, (Cooper, 2008) points out that the system is designed to be highly flexible and adaptable. Driven by rapidly changing markets and demands for time optimization, the framework has necessitated an evolution into a fifth-generation model. To design a more efficient and leaner process, tools like “value stream mapping” are used to systematically eliminate bureaucracy, wasted time, and activities that do not add real value (Cooper, 2022). Another pivotal component of the modern process is parallel processing. This concurrent approach allows activities and stages to overlap, accelerating the development timeline by enabling teams to proceed without exhaustive information, even though this entails a calculated financial risk (Cooper, 2022).

To further optimize resource allocation, the NPD process has been adapted into a highly scalable model. Because a one-size-fits-all approach is inadequate for diverse initiatives, only large and high-risk projects require the full five-stage model. Projects like product improvements or pure customer requests are deemed less complex, and teams use shortened and faster processes, for example, Stage-Gate Lite or XPress (Cooper, 2008, 2022). Simultaneously, the development process has also become much more iterative to handle projects with unclear initial needs or tech solutions. By integrating a spiral development methodology, continuous cycles are performed where the team builds, tests, gathers feedback, and refines prototypes with customers to keep designs flexible and prevent early lock-ins (Cooper, 2008, 2022).

The most significant advancement in NPD is the application of agile methodologies to physical products. This hybrid approach replaces traditional stage work with short three-to-four-week timeboxed work periods called sprints, paired with regular team alignments. At the conclusion of each sprint, a presentation takes place where concrete and noticeable results are demonstrated to stakeholders and management. Studies indicate that this agile adaptation effectively enhances flexibility, accelerates development cycles, and positively impacts team morale (Cooper, 2022). Even with the aggressive push for agile speed, keeping the strategic gates in place remains vital. Strategic gates need "teeth" to serve as strict points for actual resource allocation (Cooper, 2008, 2022).

2.4 Eco Stage Gate: Integrating sustainability and environmental goals

Pursuing sustainability provides substantial opportunities for product innovation and competitive advantage, but simultaneously it creates challenges like strict regulatory compliance. Because of this companies are reviewing their new product development (NPD) processes to better integrate sustainability principles. In response to the absence of frameworks sustainability, the conventional Stage-Gate model has been evolved into the Eco-Stage-Gate process. This adapted model prioritizes environmental goals by integrating eco-design tools, lifecycle assessments, and sustainability criteria into the process right from the start, rather than treating them as an afterthought (Cooper, 2025).

In the initial phases of development, especially during ideation, teams are highly recommended to assess and document environmental trends. Innovations designed to minimize waste generation or leverage eco-friendly materials are strongly promoted. As development

progresses to the concept stage, project teams execute initial environmental evaluations and preliminary life cycle assessments to systematically estimate the product's total ecological footprint across its entire lifespan. In the business case stage, criteria are incorporated into feasibility studies to assess energy-efficient manufacturing and supply chain sustainability. This is paired with an eco cost benefit analysis designed to weigh economic investments against the benefits of sustainable design choices (Cooper, 2025).

As the product advances into the development phase, the framework promotes sustainable design by using environmental criteria to product specifications through Design for the Environment practices. These methodologies evaluate recyclability, strategic material selection, and energy utilization. Throughout the development phase, comprehensive life cycle assessments are executed, and manufacturing processes are engineered to maximize resource efficiency and facilitate design for disassembly. In the testing and validation stage physical prototypes are subjected to field conditions to accurately estimate their energy consumption and carbon footprint, often aiming for official eco certifications from a third party. The commercialization phase strategically spotlights this sustainability achievement. The launch strategy leverages green marketing, sustainable packaging, and eco optimized distribution channels, wrapping up with a review to ensure continuous improvement (Cooper, 2025).

The execution of these stages is reinforced by utilizing AI, customized checklists, and ecocentric tools to ensure environmental metrics are handled objectively. But integrating sustainability directly into the decision gates is just as vital. To guarantee that decisions milestones are driven by objective facts rather than instinct management integrates green scorecards into the process, such as the Value-Based Scorecard. The scorecard functions as a diagnostic tool where the management can weigh a project's likelihood of market dominance and financial payoff against its integration with the company's sustainability goals. Since the scoring mechanism multiplies these variables instead of adding them, high scores elsewhere cannot counterbalance deficiencies in sustainability. Implementing such a demanding evaluation framework naturally curates a development portfolio dominated by verified sustainable innovations. Ensuring corporate authenticity and thereby substantially reducing the risk of corporate greenwashing (Cooper, 2025).

2.5 Cutting tools theory

The theory of metal cutting and the application of cutting tools represent a fundamental pillar of modern manufacturing. According to the sources, metal cutting is formally defined as a type of machining that removes unwanted material from a workpiece in the form of chips (Tooling U-SME, 2026). This process is essential because the efficiency of chip formation directly dictates the surface finish, final part quality and the overall longevity of the tool, all of which are critical drivers of manufacturing economy (Tooling U-SME, 2026).

2.5.1 Types of cutting tools

Cutting tools are categorized based on the number of cutting edges they possess. Single point cutting tools feature a single cutting edge and are primarily utilized for operations on a lathe, such as turning cylindrical parts (Tooling U-SME, 2026). In contrast, multi point cutting tools incorporate two or more cutting edges and are characteristic of tools used in milling and drilling

(Tooling U-SME, 2026). For example, a drill bit is a multi-point tool with spiral cutting edges, while a face mill uses multiple teeth to create flat surfaces on rectangular workpieces.

The core of cutting theory lies in the deformation boundary, the line where the deformed material separates from the intact workpiece as the tool applies force (Tooling U-SME, 2026). This interaction is governed by shear forces, which attempt to cause the internal structure of the material to slide against itself (Tooling U-SME, 2026). When these forces are sufficient, they produce shear strain, leading to the creation of a chip (Tooling U-SME, 2026). The physical characteristics of the chip serve as a primary indicator of whether the cutting conditions are optimized. Chips are generally categorized into two main types based on the properties of the workpiece:

- Continuous Chips: These appear as long, curled strings and are typically produced when machining ductile metals such as aluminium or copper.
- Discontinuous Chips: These fracture easily into small, separate pieces and are associated with brittle or hard metals like cast iron.

A common complication in chip formation is the built-up edge (BUE). This occurs when deformed metal adheres to the cutting edge of the tool due to high pressures and temperatures, a phenomenon most prevalent when machining ductile materials. To manage these issues, many tools incorporate chip breakers, which are designed to force chips to curl and break, ensuring efficient removal from the cutting zone (Tooling U-SME, 2026).
The efficiency of a cutting tool is heavily dependent on its geometry, specifically the rake and relief angles.

- Rake Angle: This is the angle between the front of the tool and a line perpendicular to the workpiece. A positive rake angle slopes down and away from the centerline, whereas a negative rake angle slopes up and toward the centerline.
- Relief Angle: This measures the space between the bottom of the tool and the workpiece surface. Proper relief angles are necessary to prevent the tool from rubbing against the workpiece, which would generate excessive heat and friction.

The performance of any cutting operation is dictated by three primary variables: speed, feed rate, and depth of cut. Speed generally measures the rate of rotational movement at the point of contact. The feed rate refers to the linear rate at which the tool and workpiece move relative to one another. Finally, the depth of cut is the distance the tool is plunged into the material, indicating the volume of material removed in a single pass (Tooling U-SME, 2026).

Machinability refers to the relative ease with which a material can be removed. This is not a fixed property but depends on variables like the material's hardness (resistance to penetration) and strength (resistance to deformation). Generally, materials with high hardness or strength exhibit lower machinability and cause accelerated tool wear.

Tool wear can manifest as flank wear, which is the gradual wearing of the cutting edge, or crater wear, which forms a concave depression on the tool face. If the feed rate is excessive, chipping the loss of small fragments from the tool edge may occur. Ultimately, tool life is the duration a tool functions properly before failure, and it is most significantly impacted by cutting speed.

While traditional theory focuses on mechanical tools like lathes and mills, modern research, such as the study of 17-4 PH stainless steel produced via Metal Fused Filament Fabrication (MFFF) (Molazadeh et al.,2024), highlights the integration of subtractive and additive processes. In this context, water jet machining is often categorized alongside traditional cutting as a form of machining (Tooling U-SME, 2026).

The sources note that traditional mechanical cutting can introduce stresses or inaccuracies in complex 3D printed parts. Therefore, water jet machining is utilized to extract high accuracy dog bone specimens from sintered rectangular workpieces. This method avoids the surface discontinuities and cracks that can occur during the sintering of parts with complex pre-cut geometries, ensuring that the material properties such as the longitudinal modulus of elasticity and ultimate tensile strength can be accurately measured without the interference of traditional cutting defects (Molazadeh et al., 2024).

3. Methodology

This section describes the method and approach used by the research group to carry out the thesis work.

3.1 Contact with Seco Tools

The contact person at the case company and the research group agreed that all meetings would be held digitally via email and Teams. Estifanos from the research group was responsible for acting as the contact person with the case company. In collaboration with the case company and its expectations for the thesis work, a project plan was developed. After its approval by the examiner and final revisions, the project plan was presented to the case company. Following this, discussions were held with the case company regarding the number of interviews, and which units were considered relevant to interview.

3.2 Scientific approach

The work began with the case company describing the requirements for implementing Ecodesign for Sustainable Product Regulation (ESPR), understanding and finding approaches to implement the regulation, and learning from the battery and steel industries. After the research group familiarized themselves with the topic by reading various theories on ESPR, conducting interviews in the form of a current state analysis with the sustainability team, research and development team, and product management team, along with collecting data from the case company's publicly available data on its sustainability work and pilot projects, the research group then used deductive reasoning to draw conclusions.

Deduction is a method that uses theory as a starting point to make assumptions, then compares these with empirical data to draw conclusions based on interviews by reasoning logically about the topic (Säfsten & Gustavsson, 2019).

The research method used qualitative method, was based on semi-structured interviews (8 Appendices), where several people in leading positions within the relevant teams were able to share their expertise in areas of interest to the research group.

The study was conducted as a qualitative case study, meaning that a single case in this context, a specific organization and its processes is examined to create an in-depth understanding of how sustainability work functions in practice. By focusing on a delimited case, a detailed analysis of complex relationships is made possible, although the results are primarily valid for the specific case and cannot necessarily be generalized to apply to all organizations.

3.3 Data collection

To address the purpose of the study, a combination of primary and secondary data has been used. Primary data is information that the researcher collects directly for a specific research purpose, for example through interviews, surveys or observations. Secondary data is information that has already been collected by others, for example in annual reports, official statistics reports, previous research studies or internal archival documents (Säfsten & Gustavsson, 2019).

A common and useful method for collecting primary data is the semi-structured interview. In this form of interview, the researcher has prepared a number of questions and themes to be covered in advance. But unlike a fully structured interview, the order of the questions can be changed during the conversation. The researcher also has the opportunity to ask spontaneous follow-up questions based on what the participant says (Eisenhardt, 1989). This creates a good balance between fixed structure and open flexibility. The researcher can compare answers

between different participants while at the same time unexpected and important themes can be followed up in depth.

To strengthen reliability, interviews should be recorded and transcribed verbatim. A major strength is combining primary data from interviews with secondary data, for example internal documents or official statistics. By using multiple sources, the researcher can triangulate these data, that is, cross-validate the information, which provides stronger and more credible conclusions (Gibbert et al., 2008; Säfsten & Gustavsson, 2019).

3.4 Interviews

Interviews were conducted with five candidates spanning the Product Management team, the Research and Development team and the Sustainability team. The interviews were semi structured and followed the natural flow of the conversation to allow for deeper exploration of each topic. This approach was chosen to systematically address the study's purpose and research questions concerning ESPR preparedness and learnings from the battery and steel industries. All five interviews were carried out digitally using Teams, and each session was recorded and automatically transcribed with the participants' full knowledge and consent. After the interviews were completed, relevant data was coded and compiled to create a better understanding of the subject matter, always in relation to the study's purpose and research questions.

3.5 Research ethics

At the start of each interview, the participants were informed of the four central ethical requirements for research involving human subjects. According to (Vetenskapsrådet, 2024) are information requirement, consent requirement, confidentiality requirement and usage requirement. These requirements were communicated clearly to ensure transparency and trust. The analysis of the collected data was conducted through a qualitative process where the researchers systematically searched for similarities and differences in what the informants had shared. By comparing the responses from the different departments against each other and by cross-referencing these with information from secondary sources, patterns and themes could be identified. This thematic analysis aimed to generate new insights and ideas about the investigated area rather than to measure statistical effects or frequencies.

The study has been carried out in accordance with established research ethics principles to protect the integrity of the participants beyond the source material. The work has been guided by the following four requirements. First the information requirement meant that participants were informed about the purpose of the study and the conditions of their participation. Second, the consent requirement meant that participants gave their voluntary approval to take part and were informed of their right to withdraw at any time. Third, the confidentiality requirement meant that all data was handled confidentially to protect the identity of individual individuals and departments. Fourth, the usage requirement meant that the collected information was used exclusively for research purposes.

4. Results and practical implications

4.1 ESPR in the battery and steel industries

4.1.1 Synthesized performance metrics

By combining the formal mandates of the ESPR, the Environmental Footprint methods, the GHG Protocol, and learnings from LESS and PCR 2012-02, this study identifies a possible set of measurement areas for ESPR preparedness. These should not be understood as finalized industry KPIs, but as preliminary parameters that may support further testing and discussion in cutting tool development.

Table 2: Synthesized performance metrics

Performance Parameter	**Unit of Measurement**	**Theoretical/Regulatory Basis**	**Description of application**
Energy Efficiency per Removed Volume	kWh/m³	PCR 2012:02, ESPR Annex I (g)	Measures the tool's capacity to minimize the CNC machine's energy consumption during engagement (a product-dependent process). This is determined by integrating the machine's power over the cycle time and dividing it by the volume of material removed.
Isolated Climate Impact via Delta Approach	Δ kg CO2e/m³	PEF (Delta Approach), GHG Protocol	Measures the climate advantage of a new tool against an established industry standard. The energy savings are converted into carbon dioxide equivalents, thereby isolating the tool's direct contribution to reducing the environmental impact during the use phase.
Relative Durability Indicator (Lifespan)	R_LP(X) (Ratio)	ISO 59020, ESPR Annex I (a, b)	The ratio describing the tool's expected total lifespan (including regrinding and reconditioning) divided by the industry average lifespan. It serves as empirical proof of the product's overall durability and repairability.
Circular Footprint Score	CFF-Score / %	PEF (CFF), GHG Protocol, CRM Act	Combines the proportion of recycled input material (R_1) with the proportion collected for material recycling (R_2). This approach guarantees high transparency in compliance

			with the handling requirements for critical raw materials like Tungsten and Cobalt.

The parameters in Table 2 should be interpreted as proposed measurement areas rather than fixed calculation rules. This distinction is important because cutting tools currently lack specific delegated acts and product category rules that fully define how use-phase performance should be assessed. The table therefore provides an analytical starting point for what may need to be measured, documented and verified, rather than a complete method for direct implementation. In product development, these measurement areas could be used to guide what type of evidence should be collected during testing and validation. For example, use-phase efficiency may require data on energy use, cycle time and removed material, while tool lifetime requires data on wear, tool life and possible reconditioning cycles. Product carbon footprint and circularity-related information require reliable material data, clear system boundaries and traceability between supplier information and the final product. Further practical testing is therefore needed before the proposed parameters can be used as standardized indicators.

4.1.2 ESPR In the battery industry

The function of delegated and implementing acts
Delegated acts allow the European Commission to supplement or amend non-essential elements of the primary regulation to reflect technical and scientific progress (Battery Pass consortium, 2023a). For instance, they are used to establish the precise methodology for calculating and verifying the carbon footprint (CF) of batteries, ensuring alignment with the Product Environmental Footprint (PEF) and specific category rules (Battery Pass consortium, 2023b). Implementing acts, on the other hand, are designed to ensure uniform conditions for implementation, such as defining the specific format for CF declarations or the labels required for performance classes (DIN, 2025). These acts are rolled out in stages, with EV battery requirements prioritized, followed by industrial and Light Means of Transport (LMT) batteries (Battery Pass consortium, 2023a).

Precise Implementation Criteria according to the sources, precise implementation of these acts focuses on several critical domains:

Carbon Footprint Methodology: Acts specify the functional unit as one kWh of the total energy provided by the battery over its expected service life (European Commission, 2023). They also define *hotspots,* such as cathode materials, that require company specific (primary) data rather than generic averages to prevent the underestimation of real world environmental impacts (Battery Pass consortium, 2023b).

Recycled Content and Circularity: Delegated acts establish the rules for calculating and verifying the share of cobalt, lithium, nickel, and lead recovered from waste (DIN, 2025). The sources recommend that these acts mandate separate reporting for pre-consumer waste (manufacturing scrap) and post-consumer waste (end-of life batteries) to improve traceability (Battery Pass consortium, 2023a).

Performance and Durability: For industrial and LMT batteries, acts will set mandatory minimum requirements and define technical specifications for determining the State of Health (SoH) and expected lifetime (Battery Pass consortium, 2023a).

Access Rights and Data Management: A key implementing act (scheduled for August 2026) will precisely define *legitimate interest* to determine which stakeholders, such as recyclers or second-life operators, can access specific sensitive data within the passport (DIN, 2025).

Criteria for achieving ESPR alignment

The Ecodesign for Sustainable Products Regulation (ESPR) provides an overarching framework for all Digital Product Passports (DPPs), and the battery passport is its first sector-specific application (Battery Pass consortium, 2024a). To achieve alignment with the ESPR, the battery industry must satisfy the following criteria:

Full Interoperability: The technical, semantic, and organizational aspects of the battery passport must be interoperable with other DPPs (e.g., textiles, construction) to ensure seamless data exchange across the EU market (Battery Pass consortium, 2023a).

Unique Identifiers: Alignment requires standardized unique operator identifiers and unique facility identifiers that comply with international standards like ISO/IEC 15459 to unambiguously identify value chain actors and manufacturing sites (Battery Pass consortium, 2023a).

Decentralized Data Architecture: In line with ESPR principles, data must be stored in a decentralized manner by economic operators or authorized *DPP-as-a-service* providers, with a mandatory backup copy provided to an independent service provider to ensure data persistence if a company ceases to exist (Battery Pass consortium, 2024b).

Tracking Substances of Concern: The industry must move beyond hazardous materials to track all "substances of concern" as defined by the ESPR to facilitate high-quality recycling and safe handling (Battery Pass consortium, 2023a).
Standardized Persistence: The passport system must ensure that data remains accessible for the entire expected lifetime of the product, even after the battery has been repurposed or remanufactured (Battery Pass consortium, 2023a).

4.1.3 ESPR in iron and steel industries

Iron and steel are essential materials woven into the fabric of Europe's industries and infrastructure. Producing them involves a precise, time sensitive process. It starts with raw materials mainly iron ore and scrap transformed into molten steel through either the Blast Furnace Basic Oxygen Furnace or Electric Arc Furnace methods. After secondary metallurgy, the steel is cast into shapes like slabs, blooms or billets and then hot rolled to achieve its final size. Further finishing steps such as cold rolling, annealing and coating fine tune the steel's flexibility and resistance to corrosion, making it suitable for various end uses (Arcipowska et al., 2026).

Today technical details like chemical composition and mechanical properties are typically shared via Mill Test Certificates, following the EN 10204 standard. While steel plants often rely on advanced digital systems internally, information between companies in the supply chain is still commonly exchanged as static documents, such as PDF. To improve transparency, traceability, and sustainability, the European Union has made iron and steel products a priority for the new Digital Product Passport under the Ecodesign for Sustainable Products Regulation (ESPR) (Arcipowska et al., 2026).

The DPP for intermediate steel products is designed to deliver standardized, machine readable, and interoperable data, helping drive circularity and industrial decarbonization. At the heart of

this system is the heat number, a code that links each batch of steel to its origins and production route. The digital passport will require key disclosures, such as the Product Carbon Footprint, recycled content and any Substances of Concern present in the product (Arcipowska et al., 2026).

These new requirements are meant to complement existing EU laws like the Construction Products Regulation, REACH and the Carbon Border Adjustment Mechanism. By shifting from paper based to digital data flows, the DPP could cut administrative work, streamline manufacturing and help producers benefit from the added value of greener, low emission steel. However, to fully realize these benefits, the industry must overcome hurdles such as harmonizing data methods and ensuring digital systems work smoothly across the many layers of the supply chain (Arcipowska et al., 2026).

4.2 Seco Tools and their work with sustainability

To steer measurable parameters toward strict EU legislations, we must analyze how the industry applies specific technological innovations during the use phase. Seco Tools has engineered several patented and formalized systems specifically designed to align directly with the ecodesign requirements set by Regulation (EU) 2024/1781 of the European Parliament and of the Council of 13 June 2024 establishing a framework for the setting of ecodesign requirements for sustainable products (ESPR).

One of the most prominent technologies in the turning catalogue is Jetstream Tooling. It is an advanced directed high pressure cooling technology that has been deeply embedded throughout a large segment of the tool program (Seco Tools AB, 2026b). The mechanical principle behind Jetstream Tooling involves aiming a powerful high pressure coolant stream with pinpoint accuracy directly into the vital cutting zone, positioning it exceptionally closes to the cutting edge where heat generation is most intense. The capacity of the system is highly significant. According to the turning catalogue specifications, the recommended maximum coolant pressure can achieve 275 bar (4000 psi) for square shanks with attached hoses, and 150 bar (2000 psi) for systems running a JETI adapter and a Seco-Capto connection (Seco Tools AB, 2026b). This level of extreme precision and force produces several directly measurable benefits throughout the use phase that correspond precisely with current ecodesign requirements.

Primarily, it leads to a dramatic improvement in chip control. The targeted stream physically lifts and quickly cools the chip, forcing it to break and be removed from the workspace with significantly greater efficiency compared to standard flood cooling (Seco Tools AB, 2026b). An even more critical advantage for the ESPR framework is that the technology significantly increases tool life. It radically reduces thermal and mechanical stress on the carbide by effectively removing extreme cutting heat. This prevents plastic deformation and thermal cracking to increase the durability of the tool which is a direct and mandatory requirement outlined in ESPR Annex I of the regulations.

Technology’s importance is underscored by its broad scalability across catalogue systems such as the QC modular toolholders for external turning with Jetstream Tooling (Seco Tools AB, 2026b).

To eliminate this waste the "Used Edge Detection" technology has been developed. The turning catalogue section on insert grades indicates that this highly useful function is integrated into

selected advanced products including the Duratomic series (Seco Tools AB, 2026b). In practice this technology is frequently executed using a specialized chrome-based coating on the carbide substrate designed specifically to produce an immediate and high visual contrast.

An undeniable foundation of the legislated circular economy is the extension of every single product's physical lifecycle. Within the solid end mills catalogue Solid End Mills 2026. The section dedicated to industrial services has been given a prominent placement Seco Tools AB, 2026a). To avoid the disposal of worn-sold carbide tools Seco has implemented a restoration process utilizing the same advanced technological standards employed in the original production of new tooling solutions. The tools are precisely reground to regain their original geometric specifications including the initial edge preparation and advanced surface coating. By implementing a systematized flow with ready-to-use reconditioning boxes the entire return procedure is made highly convenient for the customer. This method effectively prolongs tool life and decreases overall tooling expenses for the user while simultaneously minimizing the environmental footprint. These processes also hold global certifications under the ISO 14001 environmental management system and the ISO 9001 quality management system to ensure that all circular management practices uphold recognized industrial standards (Seco Tools AB, 2026a).

Table 3: Correlation of measurable machining parameters with EU Ecodesign requirements

ESPR parameter (Annex I and Recital 6)	**Measurable use-phase parameter**	**Technological solution / Implementation**	**Primary impact on environment and production**
Durability and reliability	Tool life, wear rate, vibration Levels	**Jetstream Tooling**: Directed high-pressure cooling (up to 275 bar) for optimized chip control.	Reduces thermal shock and plastic deformation, maximizing tool life and preventing unplanned machine downtime.
Energy and resource efficiency	Power consumption (W), torque (Nm), cutting Data (V_C, F_z)	**Cutting data optimization**: Application of specific calculation formulas and standardized data tables.	Optimized V_C and F_z eliminate unnecessary machining time, directly reducing the total energy footprint in line with EU targets.

Presence of hazardous substances	Coolant requirements (L/min)	**Precision cooling**: Controlled coolant delivery via systems such as *Precimaster Plus*.	Minimizes excess emulsion usage and reduces the volume of harmful aerosols in the working environment.
Waste generation reduction	Number of discarded cutting edges / Scrap Share	**Used Edge Detection**: Chrome-based coating technology that provides high-contrast visual wear detection.	Enables 100% utilization of carbide inserts before disposal, drastically reducing the consumption of virgin raw materials.
Reusability & refurbishment	Operational life cycles per solid tool	**Reconditioning**: Industrial process for restoring original tool geometry and coating.	Significantly extends the lifespan of high-value solid tools and eliminates the energy cost of sintering new tool bodies.
Recyclability	Recycled content of strategic materials (Tungsten/Cobalt)	**Recycling Program**: Closed-loop system for the collection and processing of used carbide tools.	Reduces dependence on critical raw materials and ensures that strategic minerals are returned to the circular economy.

4.3 Interviews from the team's triangulation

4.3.1 Product management team

Role and departmental focus

The interviewed manager leads a Product Management (PM) team that collaborates closely with R&D to define product requirements. His team specifically owns geometrical attributes, such as diameters and lengths. While his team writes the task descriptions for specific projects, broader data management and regulatory compliance often reside in specialized corporate departments.

Sustainability in development cycles

Sustainability is now a core criteria included in the task *description* for new products. Environmental requirements, such as utilizing recyclable packaging, have become mandatory. Throughout the project lifecycle, which includes five *gate* meetings, sustainability checks are performed at nearly every stage to ensure the project fulfills established environmental criteria.

ESPR and the Digital Product Passport (DPP)

The implementation of the Ecodesign for Sustainable Products Regulation (ESPR) and the Digital Product Passport (DPP) is primarily handled by the Corporate Product Management (CPM) department. This department defines the necessary attributes for items and ensures each has a corresponding value. While ESPR is identified as a potential gap in the specific PM team's current focus, the required data capabilities are distributed across CPM, Quality Assurance and Procurement.

Cross departmental collaboration

Successful product development relies on a network of specialized departments. The Project Management Office (PMO) monitors all projects and ensures all deliverables are met before passing gates. Quality Assurance (QA) manages ISO certifications, annual audits and reporting guidelines. Procurement manages interactions with suppliers, specifically regarding the purchase of raw materials like powders and tracking their traceability.

Material traceability and customer demands

Customers increasingly request detailed information regarding product coatings to ensure they do not interfere with specific workpiece materials. For example, customers may require confirmation that no titanium is present in a coating. These requests are typically addressed through safety data sheets. Regarding raw materials, Seco Tools has established a 2030 horizon for transitioning toward recyclable carbide grades.

Strategic planning and roadmaps

Product Management develops long term roadmaps to plan future investments and technological developments. These roadmaps are aligned with R&D at an early stage to identify potential *technical breaks* areas where new technology or tolerances must be developed before a project can proceed. Once aligned, the PM team creates a task description to define the project scope and requirements for R&D.

4.3.2 Research and development team

Role and desponsibility

The R&D representative is responsible for the Cut Data Engine at Seco Tools, managing a team of mechanical and IT engineers. This team develops a single-source backend service that provides data on how to use products productively, cost effectively and sustainably. While his work focuses on the use phase of products, he has limited involvement in production or material recycling specifics.

The ESPR framework and traceability

The Ecodesign for Sustainable Products Regulation (ESPR) is a European Union legislation introduced in 2024 to improve product traceability from material origin to the final product. A central element of this regulation is the Digital Product Passport (DPP), which allows for the identification of a product throughout its entire lifecycle. Implementing this regulation involves various aspects of the supply chain and production processes. Questions regarding digital threads and traceability of specific material batches were deemed outside the scope of the use phase and referred to internal experts in digitalization.

Sustainability in the use phase

Studies indicate that approximately 70% of a product's carbon footprint arises during its use phase. To manage this, Seco Tools utilizes models that factor in tool life, cycle time and cutting forces. These forces directly relate to the energy consumption required by a machine to manufacture a component. By connecting cutting data to these models, the company can perform comprehensive sustainability assessments of the use phase.

Research collaboration and transparency

Case company emphasizes a scientific foundation for its sustainability efforts to avoid *greenwashing*. The company participates in the Sentio competence center at Lund University; a five-year investment focused on sustainability and sensors. Transparency in mathematical models and data origin is considered critical for credible sustainability reporting.

Sensor technology developments

While nano sensors near the cutting edge are being researched, they currently have a low technology readiness level. This technology is primarily being driven by academia with industry support. Because these solutions may take a long time to mature, industry experts suggest that companies must act immediately on sustainability rather than waiting for sensor technology to become a complete solution.

Challenges in tool sustainability ratings

Traditional sustainability ratings, such as A-C scales, are not practical for cutting tools because the tool's impact is dependent on its specific application. A tool can only be assessed for sustainability when a specific component and the method of operation are known. The tool itself is merely an enabler to manufacture a final product, such as a brake disc, which carries the actual customer value. The goal is to determine the carbon footprint per produced component rather than the footprint of the tool itself.

Strategic optimization for carbon footprint

Sustainability in manufacturing is driven by two main factors: tool life (performance) and the carbon footprint of the tool. The production machine, such as a lathe, carries a huge carbon footprint from its initial creation and only lasts for about 30,000 hours. This means every hour it runs has a set carbon cost. Running the machine twice as fast uses twice the power, but it also finishes the part in half the time, meaning the total energy used stays the same. Producing parts faster means more components are made per hour, which spreads the machine's original carbon footprint over a larger number of finished items. To be truly sustainable, one must find an optimal operational speed, running too slowly wastes the machine's lifetime, while running too fast leads to excessive tool wear. Fortunately, sustainability often aligns with productivity and cost effectiveness, allowing manufacturers to achieve all three without necessarily sacrificing one for the other.

The R&D representative reported that the team has developed mathematical models that connect cutting data with the sustainability performance of the use phase. The available model components include estimates of tool life and wear, cycle time, cutting forces, energy demand and material losses associated with tool wear. According to the interview, parts of the mathematical and calculation approach used in the calculation engine were patented in the early 2020s. The representative also stated that the models and their underlying mathematics were being prepared for publication in collaboration with Lund University.

4.3.3 Sustainability team

Results from the sustainability expert interview: Strategy, data management and use phase
To comprehend how legislation and theoretical sustainability frameworks are applied in practice an interview was conducted with a global sustainability expert at Seco Tools. The interview focused on the company's adaptation to the ESPR, its implementation of digital product passports, and the challenges related to life cycle assessments. The results of the interview were organized into four key themes.

Strategic adaptation and uncertainty surrounding the ESPR

Regarding the implementation of the ESPR, the current situation is characterized by considerable uncertainty concerning the extent to which the requirements will apply to consumable products such as cutting tools. The sustainability expert explained that the company is taking a proactive approach by participating in pilot projects with Swedish industrial partners to explore how a theoretical DPP could function in practice. The objective is to attain regulatory alignment for steel tool holders by 2027 although the implementation trajectory for carbides is anticipated to extend until 2028 or later due to inherent material complexities. The initiative is managed centrally by a specialized sustainability team working closely with R&D and product management, with strong backing from executive leadership since sustainable manufacturing is considered a key enabler of the company's 2030 goals.

Data management and challenges within the supply chain

The most significant challenge in meeting the requirements of the ESPR and DPP lies not primarily in the absence of primary data, but in the complexities of data management. The company has a portfolio consisting of over 30,000 standard products in addition to tens of thousands of customized special tools. Although data concerning energy, water, and gas usage is generally accessible, it is distributed across multiple internal systems such as production

systems, ERP systems and product information systems. The primary difficulty resides in the extraction and structuring of this information to achieve a granular level of detail for each individual insert or tool.

Regarding the supply chain there is inconsistency in the quality of data provided. The organization possesses direct access to primary data from several major steel suppliers as well as for key raw materials such as tungsten carbide and cobalt. For other materials where the company is a minor customer, materials such as tantalum. It becomes more difficult to request specific environmental data, which forces reliance on secondary databases like Ecoinvent.

The Sustainability Expert identified raw materials, sintering and coating, as the principal contributors to the carbon footprint of a carbide insert. In comparison, secondary manufacturing processes such as grinding, edge honing and cleaning were estimated to account for approximately two per cent or less of the product footprint. The interview indicated that the relative contribution of these processes is currently being considered when discussing the appropriate level of detail and the possible use of average data in product-level carbon-footprint calculations.

Methodological challenges in LCA and the use phase

Due to a lack of an industry specific product category rules (PCR) for cutting tools, Seco Tools relies extensively on general ISO standards such as ISO 14040 and 14044 for life cycle assessments alongside ISO 14001 and ISO 50001. The interview highlights a central and critical challenge in calculating the environmental impact of tools during the use phase namely the selection of an appropriate functional unit. According to the sustainability expert, internal tests have demonstrated that LCA results can change dramatically and even reverse depending on the parameter used. Measuring environmental impact in terms of removed material yields a different outcome compared to using material removal rate as the basis of assessment. A longer tool lifespan is generally considered positive for sustainability but may appear negative in a carbon perspective if an inappropriate functional unit is used. In response to this issue, the company has implemented an application specific approach that relies on its internal cut data engine instead of generic cradle-to-gate averages.

Competitiveness and the balance between sustainability and cost

The interview indicates that Seco Tools together with the Sandvik Group maintains a pioneering position within the sector concerning the development of LCAs and the implementation of DPP trials. The only identified competitor operating within this domain employs a simplified weight based averaging model whereas Seco Tools is developing a significantly more granular methodology. Ultimately the research indicates that transitioning to sustainable production inputs like green steel or specialized coatings necessitates a financial premium that increases the overall production cost per tool. Despite these costs there is clear market receptivity toward premium pricing provided that the organization can precisely document and prove a significant reduction in environmental footprint. This makes life cycle assessments and data management an essential business requirement for future product development.

4.3.4 Sustainability compliance officer, insights in DPP

Interpretation of ESPR and managing uncertainty

The Ecodesign for Sustainable Products Regulation (ESPR) is currently viewed as a generic, high level framework rather than a set of specific instructions. Because the regulation announces requirements that are yet to be finalized, the focus is on preparing for future uncertainty. This preparation involves leveraging existing knowledge from other sustainability regulations and aligning with customer interests to anticipate upcoming demands.

The Digital Product Passport (DPP) pilot project

Seco Tools is currently engaged in a pilot project involving both internal and external actors to prepare products for the Digital Product Passport (DPP). This pilot follows a lifecycle approach, starting with a DPP from an intermediate steel maker (the supplier), incorporating it into Seco's manufacturing process and finally delivering a DPP to the customer. The project is structured into specific work packages to manage the data flow from the supplier through to the end consumer.

Organizational responsibilities and frameworks

The implementation of sustainability requirements is divided across different functions. While some team members focus on the DPP specifically, others are responsible for building a broader framework for sustainable products. Support is also provided to procurement for collecting supply data required by existing regulations. This collaborative approach ensures that various aspects of product compliance and sustainability are addressed by specialized colleagues.

Lessons from the battery and steel industries

Although Seco does not produce batteries, the company shares information with other Sandvik entities that do. The battery industry is already working intensively on mapping data and implementing the necessary IT structures to meet a January deadline. These industries provide a benchmark for how to handle high level information sharing and the challenges of disclosing specific data points.

Data points and standard selection

To avoid *overdoing* work before official delegated acts are released, Seco Tools focuses on basic data points rather than trying to meet every possible ISO standard. Current efforts center on compliance, material origin, supplier details, recycling content and carbon footprint (LCA). Regulated substances and material composition are also prioritized as fundamental components of the necessary data set.

The challenge of regulated substances and traceability

The Sustainability Compliance Officer identified regulated substances as one of the most significant current data gaps. Information may originate from suppliers, internal manufacturing or products that are purchased and subsequently resold, but it is not consistently connected to the relevant product records. For example, a supplier may report that a component is compliant or does not contain regulated substances such as lead or PFAS. However, the company's current information systems cannot always connect that supplier declaration to the specific component and subsequently identify the final tool in which the component was used. The interviewee illustrated this problem through the difficulty of determining in which finished tool a particular purchased screw had been incorporated. This missing connection between supplier

declarations, components and finished products represents a major challenge for product-level substance traceability and future Digital Product Passport readiness. The interview also identified clearer data ownership and responsibility within Product Management and R&D as necessary for ensuring that regulated-substance information is entered, maintained and linked to the relevant products.

Competitive positioning and benchmarking

Seco is considered an early mover and is *very ahead* regarding DPP and ESPR piloting, often presenting its progress to industry peers. However, the company may be behind sectors like the automotive industry, where companies such as Volvo already have advanced systems for identifying all product substances. The company's standing varies by topic, it leads in LCA and service offerings but acknowledges room for improvement in human rights reporting and chemical tracking.

Future outlook and regulatory monitoring

The company is currently waiting for delegated acts to determine if all finished products will eventually require a DPP or if the focus will remain on intermediate products. To stay informed, Seco monitors scientific studies and data point proposals from the Joint Research Centre (JRC). This proactive monitoring allows the company to identify gaps and maintain readiness without prematurely committing to unverified requirements.

4.3.5 Sustainability and EHS Council Execution Lead

New product sustainability framework

Seco is developing a framework where sustainable product development is based on Life Cycle Analysis methodology. This framework aims to establish environmental baselines for products currently in the development roadmap. The shift moves the organization away from the previous Sustainability Impact on Customers and Operations model (SIC and CO), which was more prescriptive, toward a cradle-to-gate LCA approach.

Integration into the gate decision model

The company is currently integrating sustainability criteria into its gate decision model for new product development. This process involves collaboration between Product Management (PM) and Research and Development (R&D) teams. Once finalized, these sustainability checks will become the standard norm for compliance throughout the product development lifecycle.

Cradle to gate vs use phase impact

While approximately 60% to 70% of a cutting tool's environmental impact occurs during the use phase, conducting a full cradle-to-grave LCA is challenging because application variables, such as machine health and operator knowledge are outside Seco's control. Consequently, the company focuses on cradle-to-gate data, providing customers with reliable figures for kg CO2 equivalent per tool.

Sustainability KPIs and tool longevity

Beyond carbon footprint, Seco is identifying meaningful sustainability Key Performance Indicators (KPIs) beneficial to customers, such as tool life extension. The company is exploring designs that allow for multiple life cycles, such as using additive manufacturing to increase the number of possible regrinds from 3 to 16.

Organizational maturity and pilot projects

Awareness of the Ecodesign for Sustainable Products Regulation (ESPR) and Digital Product Passports (DPP) varies across different functions. While IT and sustainability teams lead these efforts, the knowledge transfer to other departments is an ongoing process. LCA pilot projects are currently active in specific Seco locations to ensure readiness before delegated acts take effect, likely between 2029 and 2030.

Supplier traceability and data gaps

Obtaining detailed traceability documentation from suppliers remains a challenge and is currently considered hypothetical. Most suppliers, including steel manufacturers, are only beginning to explore DPP requirements and do not yet provide QR codes or automated traceability data for their products.

Internal value chain and raw materials

As part of the Sandvik Group, Seco tools value chain includes internal suppliers like Wolfram (WBH) for tungsten carbide powder. Seco is pushing these upstream entities to provide batch traceability and LCA data. Critical raw materials being monitored within this internal and external supply chain include steel, tungsten carbide and cobalt. The Sustainability and EHS Council Execution Lead further explained that the physical form of tungsten carbide powder limits the achievable level of product-specific traceability. For this material flow, batch-level traceability is currently considered more technically feasible than tracing the powder to an individual final product.

5. Analysis and discussion

5.1 ESPR in the battery and steel industry

The battery industry illustrates how regulation can compel companies to move from broad sustainability reporting to product-specific, machine-readable data. For cutting tools, the strongest transferable lesson is that a DPP must be designed as an information system based on verified data fields, rather than as a label intended for customers. This means that material origin, recycled content, substances of concern, product identifiers, supplier declarations, and carbon footprint data should have defined owners and verification routines already during development.

By comparing the battery industry's requirement for a decentralized data architecture with Seco Tools' current situation, it becomes evident that Seco's fragmented data across multiple ERP and production systems will be a major hurdle for DPP implementation. By adopting the battery sector's established protocols for data access, Seco can create a robust mechanism to manage sensitive supply chain details while simultaneously improving the reliability of data from minor suppliers.

The steel industry contributes to a different but equally important lesson. The LESS framework and the steel decarbonisation scale show that absolute carbon footprint values can be misleading when products differ in material route, alloy content and technical complexity. A high alloy or high-performance material may have a larger absolute footprint because it performs a function that simpler materials cannot. This insight is central for cutting tools. A carbide insert with a higher cradle-to-gate footprint may still reduce total environmental impact if it extends tool life, lowers cutting energy, improves surface quality or reduces rejected components during use.

This means that direct transplantation of battery or steel indicators to cutting tools would be insufficient. Instead, the indicators must be translated. Battery logic should guide how data are collected, structured and verified. Steel logic should guide how performance is evaluated fairly, especially when recycled content, alloying elements and technical performance create trade-offs. The combined lesson is that cutting tools sustainability should be measured through both product data and application-based performance.

5.2 Resolving the functional unit paradox via PEF

Internal empirical testing has demonstrated that the selection of an inappropriate functional unit can cause LCA data to fluctuate dramatically, culminating in diametrically opposed conclusions. For example, evaluating environmental impact based strictly on the total volume of removed material yields a completely different carbon profile than an assessment based on the material removal rate. A major paradox emerges regarding tool longevity. While extending a tool's lifespan is ideal for saving resources, it can show up as a negative outcome for your carbon footprint depending on the measurement unit you use. To resolve this methodological uncertainty, organizations have engineered customized approaches. Instead of relying on generic industry averages, they use highly detailed internal cutting data to accurately model how the tool behaves during use.

Internal analysis at Seco Tools demonstrates a significant risk in current sustainability reporting, as the selection of a functional unit can lead to contradictory Life Cycle Assessment results. This challenge highlights that environmental claims may be dismissed as subjective or

potentially deceptive without the PEF's Delta approach, which provides the necessary framework to isolate product-dependent energy data during the machining process.

The PEF methodology provides two distinct approaches for allocating environmental impacts during the use phase. The Main Function approach attributes all environmental impacts directly to the product system. However, the Delta approach is applied when a product's interaction with a larger system causes an increase or decrease in the consumption of related resources. For example, if a substandard cutting tool generates excessive material waste or draws excessive electricity due to high friction during machining, the Delta approach establishes a reference consumption baseline. The resulting excess environmental footprint is then allocated directly to the substandard product responsible for creating systemic inefficiency.

A critical finding from both the literature and the interviews is that the environmental performance of cutting tools cannot be evaluated independently from the manufacturing application in which they are used. Internal testing at Seco Tools demonstrates that different functional units generate different LCA outcomes, which can lead to contradictory conclusions regarding the most sustainable course of action. This creates a fundamental challenge for future ESPR implementation because the regulation seeks standardized and comparable sustainability metrics while the practical reality of machining is highly application dependent.

The Product Environmental Footprint methodology helps address this issue by implementing stricter rules for modelling and more consistent principles for allocation. Assigning a single carbon footprint value to a cutting tool is not particularly useful. Because most of the environmental impact happens while the tool is in use, the actual footprint depends almost entirely on how the machine is set up, how the tool is operated, and the specific choices made during the manufacturing process. As a result, future ESPR guidelines for cutting tools will likely need to rely on application-specific metrics rather than generalized averages to be truly effective.

This observation makes a direct contribution to answering Research Question 1. The analysis suggests that future ESPR requirements for cutting tools will focus not on product carbon footprints alone, but on methodologies capable of accounting for use-phase variability. Companies that possess advanced modelling capabilities and extensive machining data will secure a significant competitive advantage.

5.3 Overcoming scrap scarcity with LESS adjustments

The evaluation of environmental performance in manufacturing is further complicated by physical limitations within material supply chains. The Low Emission Steel Standard (LESS) and its associated Steel Decarbonization Scale provide an example of how carbon intensity, production routes, scrap share and material complexity can be considered within an environmental classification framework. A regulatory framework or ecolabel that evaluates sustainability strictly based on an absolute Product Carbon Footprint may generate distorted comparisons if the limited availability of recycled materials is not considered. The LESS framework demonstrates that environmental performance cannot be assessed only by rewarding a high scrap share, since the amount of available post-consumer scrap is physically constrained by historical production volumes and the long service life of steel products. Focusing just on

recycled content might encourage competition for the same limited amount of scrap without necessarily creating an equivalent reduction in total environmental impact.

Within the LESS framework, the permitted emission thresholds decrease as the scrap share increases. An increased proportion of recycled content does not guarantee an easier path to a high environmental classification, because production processes that consume massive scrap figures must meet far tougher carbon constraints. This mechanism illustrates the importance of evaluating both material inputs and the actual efficiency of the production process. Seco Tools' recycling and reconditioning initiatives support resource efficiency through different mechanisms. The recycling program returns valuable materials to the material loop and may reduce the need for primary raw material inputs. Reconditioning, by contrast, extends the operational life of an existing tool and delays the need to manufacture a replacement. Both activities support circularity, but reconditioning should not automatically be interpreted as increasing the recycled content of a new cutting tool.

The Sustainability expert stated that transitioning to more sustainable production inputs may increase overall production costs, though customers might accept a premium price when the reduction in environmental impact can be documented and verified. An adaptation standard modeled after LESS for cutting tools might therefore promote a more balanced evaluation of production and material related climate impacts by accounting for necessary technical complexity. That said, such an adjustment would not by itself prove superior to environmental performance during the use phase. Use phase advantages must instead be demonstrated through parameters such as tool life, cutting energy, cycle time, cutting forces and material losses.

The LESS framework should therefore be interpreted as an analytical precedent rather than as a calculation method that can be transferred directly to cutting tools. The LESS framework adjusts environmental metrics based on the production route, scrap share, and material complexity of steel. whereas the interview findings show that the environmental performance of a cutting tool is also strongly dependent on its application. Recycled content and cradle-to-gate emissions should therefore not be treated as isolated decision criteria. They need to be evaluated together with tool life, cutting energy, cycle time and material losses during machining. This combined evaluation ensures that a tool featuring high recycled content, but poor use phase performance prevents us from automatically being classified as the more sustainable alternative. This balanced evaluation ensures that advanced cutting tools from being penalized just because the material composition required to achieve its functional performance creates a higher initial carbon footprint.

5.4 Digital Product Passports and data management complexities

To comprehend how these theoretical concepts and legislative rules overlap in practice, we can analyze data from corporate sustainability experts. This information highlights the profound complexities involved in actual strategy and data management. The industrial sector is currently unsure about how ESPR regulations will apply to consumable and semi-consumable parts. This is because core components like tool holders can last for years, whereas individual carbide inserts might degrade within weeks.

To address regulatory uncertainty, Seco Tools is proactively participating in cross-industrial Digital Product Passport pilot projects. They want to make sure their steel holders meet all standards by 2027, and their complex carbide products do the same by 2028. Overseeing a portfolio of more than 30,000 products, the company finds it hard to gather basic environmental facts because they are scattered across different ERP and manufacturing systems. This makes

it tough to create detailed profiles for individual tools, in addition, gaps in supply chain data create a massive roadblock. While Seco Tools holds primary data for main materials like steel, cobalt, and tungsten, it lacks the market leverage to demand primary data for minor materials like tantalum. This forces the company to use general databases like Ecoinvent, which naturally makes the final Life Cycle Assessment less accurate.

The empirical findings demonstrate that the core DPP challenge is not just an absence of environmental information, but the failure to maintain a continuous connection between supplier information and the final product. A supplier may confirm that a particular component does not contain regulated substances such as PFAS or lead, while the internal systems may still be unable to identify the exact finished tool in which that component was used. The information exists, but its connection to the final product becomes lost as the material progresses through the internal value chain.

The traceability challenge also differs depending on the type of material. Provided the identifier is preserved throughout production, a supplier declaration can connect to a specific batch or item for purchased components. Because tungsten carbide powder and other materials are mixed and processed in larger material flows, batch level traceability is more practical than tracing the material to each individual finished tool. The findings thus indicate that DPP information must draw a strict distinction between product specific data, batch specific data, and values obtained from secondary databases. The main implementation barrier for Seco Tools does not rest just on the creation of the passport interface, but rather on building a stable digital thread across suppliers, internal production systems, ERP systems, product information systems, and finished cutting tools. This digital thread is necessary to ensure that environmental and compliance information remains connected to the relevant product throughout its lifecycle.

5.5 Learnings from the ESPR regulation

5.5.1 Technological manifestations of ESPR parameters

To translate the broad objectives of the ESPR into actionable regulations, the European Commission utilizes a predetermined set of product parameters to form the technical foundation for environmental performance. A critical analysis reveals a profound overlap between these theoretical parameters and the specific technological innovations developed within the cutting tool sector.

The formal ESPR parameter of durability and reliability demands that a product's guaranteed lifetime and technical durability be explicitly defined, incorporating the mean time between failures and structural resilience under operational load. For industrial machining applications, this abstract requirement corresponds directly to measurable parameters including tool life, wear rate, and vibration levels. A clear manifestation of this technological implementation is found in systems like Jetstream Tooling, which uses specialized, directed high-pressure cooling mechanisms across a broad range of turning applications. From an environmental and operational perspective, the primary benefit of this technological advancement lies in its mitigation of thermal shock and plastic deformation. By actively cooling the cutting interface, the technology drastically extends the tool's lifespan and prevents unplanned machine downtime, thereby directly satisfying the ESPR durability objectives. In a similar manner, the

resource use parameter within the ESPR framework targets the consumption of power, water, and strategic raw materials across the entire product lifecycle, focusing heavily on the physical variables that determine absolute efficiency. The practical overlap here is quantified through measurable use-phase parameters, including power consumption expressed in watts (W), torque in newton-meters (Nm), and highly specific cutting data variations such as cutting speed and feed rate. From an operational perspective, implementation depends on cutting data optimization using industry standard data tables and specific calculation formulas. By successfully optimizing these machining variables, organizations can eliminate redundant processing cycle times, which in turn systematically lowers the aggregate power consumption of CNC machine tools to satisfy European Union resource use targets.

The regulation also outlines the presence of substances of concern, establishing criteria to avoid technical solutions that prevent downstream recycling efficiency or amplify environmental footprints. In operational environments, this regulatory parameter translates into the continuous tracking of coolant flow demands measured in liters per minute (L/min). Through the integration of controlled coolant delivery mechanisms such as the Precimaster Plus system, the volume of excess emulsion used during chip formation is minimized. As a result, the emission of hazardous airborne aerosols into the immediate manufacturing zone is significantly reduced, ensuring the production facility remains fully compliant with hazardous substance tracking protocols.

Taking this step further, the environmental parameter centering on waste generation reduction requires organizations to track and minimize the physical disposal of materials. From a metric perspective, the industry tracks this progress by measuring the absolute number of spent cutting edges and monitoring fluctuations in the total scrap ratio. Meeting this technical challenge head-on is the introduction of Used Edge Detection, a patented innovation built directly into advanced product lines like the Duratomic series. This specific capability is achieved by applying a specialized chrome-based coating over the carbide foundation, which creates a sharp, highly visible color shift at the first sign of abrasion. The practical outcome is the absolute elimination of premature disposal, enabling operators to run every carbide insert to its true operational limit, thereby minimizing scrap rates and drastically driving down the consumption of virgin raw materials.

The theoretical demands for reusability and upgradability, as well as repairability and maintenance, evaluate a product's potential for nondestructive dismantling, refurbishment, and reliance on standardized components. The practical overlap within the cutting tool sector is addressed through established reconditioning services featured within the solid end mills catalog. This process requires precise tool regrinding, resetting the original cutting-edge shapes, and reapplying advanced surface coatings. Supported by organized workflows and ready-to-use return boxes, this service extends the working life of each solid tool and avoids the high energy costs of manufacturing entirely new tool bodies.

The primary method used to evaluate and measure these environmental footprints is the Life Cycle Assessment (LCA). The formalized ISO 14040 standard defines an LCA as a structured review of all inputs, outputs, and environmental impacts of a product system. It tracks the

complete journey of materials from raw extraction all the way to final disposal. The structured LCA framework functions as a four-phase cycle. This involves defining the system scope, auditing material and energy inventory data, evaluating environmental impact categories, and interpreting the final metrics to drive sustainability strategy. Therefore,

5.5.2 Integrating ESPR into the product development process

To prepare for the ESPR, Seco Tools needs to update its product development process. Table 4 illustrates this transition by combining theory with our interview results. The table clarifies the sustainability focus, current challenges, and the required data at each development gate.

Table 4: ESPR integration in the product development process

Development Phase	Traditional focus	Sustainability focus under ESPR	Main challenge
Gate 1: Ideation	Product idea, customer needs, technical possibilities	Early evaluation of recycled materials and critical raw materials	Difficulty in setting sustainability targets as future ESPR requirements are unclear
Gate 2: Business case	Costs, profitability, project planning	Early carbon footprint estimation and supplier mapping	Hard to predict future environmental requirements and standards
Gate 3: Development	Prototyping and tool performance testing	Collection of LCA data and evaluation of recyclable design	Supplier environmental data is often incomplete during development
Gate 4: Validation	Tool life tests and machining validation	Verification of environmental performance and DPP preparation	Sustainability data is spread across several systems and suppliers
Gate 5: Commercialization	Product launch, sales and marketing	Final DPP implementation and recycling preparation	Sustainability information must be communicated clearly without creating misleading claims

The empirical findings in Table 4 show a clear shift from standard New Product Development toward the Eco-Stage-Gate framework proposed by Cooper (2025). While traditional product development within the cutting tool sector strictly prioritized technical performance and cost efficiency, the looming ESPR mandates an integration of environmental criteria at every decision gate. As confirmed by a lead expert in sustainability, Seco Tools now embeds checks for sustainability inside gate decision models to ensure compliance across the

product life span. This strategic shift aligns with de Moraes et al. (2025), who emphasize that integrating LCA and Ecodesign in the early phases of development is critical for strategically optimizing environmental metrics without sacrificing product quality.

However, the empirical data reveals significant practical friction when applying the theoretical Eco-Stage-Gate model to the cutting tool industry. According to Cooper's (2024) framework, early stages (Gate 1 and 2) should involve preliminary LCAs and supplier sustainability mapping. As highlighted by the Sustainability Expert and the Sustainability Compliance Officer, early environmental data mapping faces a major hurdle due to supply chain depth and data fragmentation across internal systems. While test centers offer a fast way to measure tool performance, tracing regulated substances or exact recycled content from upstream suppliers such as Tantalum during the early development gates is currently described as a "data gap" or "hypothetical". This indicates that while the structure of an Eco-Stage-Gate exists, the data maturity required to pass these gates with total assurance is still under development.

The modern stage-gate process rests on the idea that strategic gates must have teeth to serve as strict points of control for the use of resources (Cooper, 2008, 2022). The interviews with Product Management confirm that sustainability has evolved from a post-launch data point into an essential condition for progress. If a product fails to meet the established environmental criteria, such as recyclability or carbon footprint, the project cannot pass the gate. The path to achieving a competitive advantage under the ESPR will not solely depend on the physical performance of the tools, but on overcoming the digital complexities of data management to satisfy the strict demands of the Eco-Stage-Gate process.

5.5.3 Critical discussion and limitations

In the absence of a defined PCR for cutting tools, the recommended functional units and allocation rules should be viewed as a preparedness framework rather than a final compliance method. The use phase remains highly dependent on factors beyond the control of the tool manufacturer, including the electricity mix, coolant strategy, workpiece material, fixture stability, operator competence, and machine condition. This requirement highlights the importance of differentiating between standardized test centre results and application-specific performance data collected from customers.

A notable limitation is that the empirical findings are derived from interviews within one specific company. While this methodology ensures high practical relevance and depth, it also means that the conclusions are heavily influenced by the internal organizational structure, strategic focus, and data maturity of Seco Tools. Given this scope, the conclusions should not be extended to all cutting tools manufacturers until further comparative research can validate these findings across a wider range of companies. The primary advantage of the case study methodology is its capacity to identify specific implementation barriers that a strictly theoretical analysis might overlook.

A deeper analytical limitation highlighted by the interviews is the necessary reliance on cradle-to-gate LCA data due to the unpredictability of the use phase. While Seco provides reliable kg CO2e data up to the factory gate, this approach inherently ignores the 60-70% of the environmental impact that occurs during machining. The resulting gap between theoretical metrics such as the Delta Approach and real-world application, there is an urgent need for industry wide Product Category Rules designed specifically to address the unique complexities of cutting tools.

6. Conclusion

This thesis sets out to investigate how the upcoming Ecodesign for Sustainable Products Regulation (ESPR) impacts the development of cutting tools from a life cycle perspective, using precedents from the battery and steel industries. The work was guided by two research questions, which are answered below.

6.1 Answer to research question 1

What requirements do ESPR and associated standards impose on cutting tools from a life cycle and Ecodesign perspective, based on learnings from the battery and steel industries?

The analysis shows that the ESPR does not impose a fixed list of requirements on cutting tools today. Instead, it establishes a framework of product parameters covering durability, repairability, recyclability, recycled content, resource use and presence of substances of concern. These parameters will be specified through future delegated acts European Parliament and Council (2024b). For cutting tools, the lack of finalized Product Category Rules means that specific requirements remain undefined for now Faraca et al. (2024). This regulatory uncertainty was confirmed by the Sustainability team at Seco Tools, who explained that the company is taking a proactive approach by participating in pilot projects with Swedish industrial partners to explore how a theoretical Digital Product Passport could function in practice per the interview with the Sustainability expert. The Sustainability Compliance Officer added that the ESPR is currently viewed as a generic, high-level framework rather than a set of specific instructions, which means the focus is on preparing for future uncertainty rather than implementing fixed rules states the Sustainability Compliance Officer.

Looking at the battery industry first, the key requirement is that the Digital Product Passport must function as a verified, machine-readable information system rather than a customer facing label. This system demands decentralized data architecture, unique product identifiers, and mandatory data persistence (Battery Pass consortium, 2023a) and (Battery Pass consortium, 2024b). For cutting tools, this means that companies need to establish ownership and verification routines for material origin, recycled content, substances of concern and carbon footprint data during product development, not after the fact (Battery Pass consortium, 2023a). The Sustainability and EHS Council Execution Lead confirmed that Seco Tools is currently integrating sustainability criteria into its gate decision model for new product development, which involves close collaboration between Product Management and R&D teams as stated by the Sustainability and EHS Council Execution Lead interview. However, the same interview revealed that awareness of the ESPR and Digital Product Passports varies across different functions and knowledge transfer to other departments is an ongoing process.

From the steel industry, specifically the Low Emission Steel Standard and the steel decarbonization scale, the key lesson is that absolute carbon footprint values can be misleading when products differ in material complexity and alloy content. The requirement is therefore to evaluate performance fairly, using adjustment rules for quality steel and high-performance materials (LESS aisbl, 2025a) and (LESS aisbl, 2025b). For cutting tools, this means that a carbide insert with a higher cradle to gate footprint might still be more sustainable if it extends tool life or reduces cutting energy (ResponsibleSteel and LESS, 2025a). The R&D team at Seco Tools confirmed this reasoning, stating that sustainability in manufacturing is driven by two main factors, tool life and the carbon footprint of the tool. Running a machine twice as fast uses twice the power, but it also finishes the part in half the time, meaning the total energy use stays the same. The expert further explained that to be truly sustainable, one must find an optimal

operational speed because running too slowly wastes the machine's lifetime, while running too fast leads to excessive tool wear according to the interview with the R&D representative.

When synthesizing these insights, three key areas of ESPR preparedness can be identified for cutting tools. First, companies need product specific, verifiable data fields for material traceability and carbon footprint. Second, they need decentralized and interoperable Digital Product Passport architecture. Third, they need a fair evaluation methodology that accounts for use phase performance rather than penalizing high alloy, high-performance products.

6.2 Answer to research question 2

What measurable use phase product performance parameters can be applied in the product development of cutting tools?

Based on the theoretical integration of the ESPR, Environmental Footprint methods, life cycle assessment and the LESS framework, supported by the empirical findings from the interviews, the study identifies four proposed measurement areas for ESPR preparedness in cutting tool development. These areas are use-phase efficiency, climate impact, tool lifetime and circularity-related material data. They should be understood as preliminary measurement areas rather than fully validated KPIs, since delegated acts and product-specific calculation rules for cutting tools have not yet been established.

The first parameter is Energy Efficiency per Removed Volume, measured in kilowatt hours per cubic meter. This parameter measures the tool's ability to minimize the CNC machine's energy consumption during cutting. It can be verified by integrating the machine's power draw over the cycle time and dividing that number by the volume of material removed. This parameter directly addresses the ESPR requirement for resource efficiency (European Parliament and Council, 2024b). The R&D team confirmed that cutting forces directly relate to the energy consumption required by a machine to manufacture a component and by connecting cutting data to internal models, the company can perform comprehensive sustainability assessments of the use phase stated the R&D representative in the interview.

The second parameter is Isolated Climate Impact via the Delta Approach, measured in kilograms of carbon dioxide equivalent per cubic meter. This parameter measures the climate advantage of a new tool compared to an established industry standard. By converting energy savings into carbon dioxide equivalents, the tool's direct contribution to reducing environmental impact during the use phase becomes clear. This approach resolves the functional unit paradox that emerged in our interviews. The Sustainability team explicitly stated that internal tests have demonstrated that Life Cycle Assessment results can change dramatically and even reverse depending on the parameter used. Measuring environmental impact in terms of removed material yields a different outcome compared to using material removal rate as the basis of assessment says the Sustainability Expert in the interview. The R&D team added that a tool can only be assessed for sustainability when a specific component and the method of operation are known, which is why the goal should be to determine the carbon footprint per produced component rather than the footprint of the tool itself stated the R&D representative in the interview.

The third parameter is the Relative Durability Indicator. This is a ratio that describes the tool's expected total lifespan, including regrinding and reconditioning, divided by the industry average lifespan. This indicator serves as empirical proof of the product's durability and repairability, directly addressing the ESPR requirements for durability and reliability (European Parliament and Council, 2024b) and (ISO 14044:2006). The Sustainability and EHS Council

Execution Lead confirmed that beyond carbon footprint, Seco Tools is identifying meaningful sustainability Key Performance Indicators that benefit customers, such as tool life extension. The company is exploring designs that allow for multiple life cycles, for instance using additive manufacturing to increase the number of possible regrinds from three to sixteen according to the interview with the Sustainability and EHS Council Execution Lead.

The fourth parameter is the Circular Footprint Score, expressed as a percentage. This parameter combines the proportion of recycled input material with the proportion collected for material recycling. It ensures high transparency in compliance with the Critical Raw Materials Act, particularly for strategic materials like tungsten and cobalt (European Parliament and Council, 2024a). This parameter addresses both the recycled content and resource use requirements of the ESPR. However, the interviews revealed significant challenges in this area. The Sustainability Compliance Officer stated that the greatest challenge and most significant data gap involves regulated substances. There is currently no centralized system to track the presence of substances such as cobalt, nickel, lead or PFAS in every product and it is difficult to connect specific supplier compliance data to individual tools or components according to the interview with the Sustainability Compliance Officer. The Sustainability and EHS Council Execution Lead added that obtaining detailed traceability documentation from suppliers remains a challenge, as most suppliers are only beginning to explore Digital Product Passport requirements.

These four parameters are not just theoretical ideas. They can be verified in test centers using existing Seco Tools technologies such as Jetstream Tooling for energy efficiency and tool life, as well as reconditioning services for durability and circularity (Seco Tools AB, 2026a) and (Seco Tools AB, 2026b). The practical implication is that Seco Tools can integrate these metrics into its Stage Gate development model, particularly at the development and validation stages, to verify environmental performance before a product is launched (Cooper, 2008) and (Cooper, 2025). The Product Management team confirmed that throughout the project lifecycle, which includes five gate meetings, sustainability checks are performed at nearly every stage to ensure the project fulfills established environmental criteria.

6.3 Theoretical contributions

This study makes four theoretical contributions. First, it synthesizes four measurable use phase performance parameters for cutting tools by integrating ESPR, PEF Delta, ISO, and LESS frameworks. Second, it resolves the functional unit paradox in cutting tool Life Cycle Assessment by applying the Delta approach to isolate product dependent impacts. Third, it adapts the LESS steel decarbonization scale logic to cutting tools, arguing that higher cradle to gate footprints can be justified by superior use phase performance. Fourth, it extends Eco Stage Gate theory by specifying how ESPR requirements can be operationalized at each development gate under regulatory uncertainty. These contributions bridge EU regulation and product development theory.

6.4 Practical contributions

This study offers several practical contributions for cutting tool manufacturers. First, it proposes four measurement areas that can support ESPR preparedness in cutting tool development, use-phase efficiency, climate impact, tool lifetime and circularity-related material data. These areas can guide what data companies may need to collect and verify. Second, it demonstrates how existing technologies such as Jetstream Tooling and reconditioning services can be leveraged to verify these parameters in test centers. Third, it offers a cross-sectoral learning framework, drawing practical lessons from battery and steel industries on Digital Product Passport

architecture and fair carbon evaluation. Fourth, it provides a roadmap for integrating sustainability checks into each Stage Gate decision point, enabling proactive ESPR compliance before delegated acts are finalized.

6.5 Limitations of the study

The study has several limitations. First, it is based on a single case company, Seco Tools, which limits the generalizability of the findings to the entire cutting tool industry. Second, the study was time constrained to ten weeks, which limited the opportunity to further investigate and validate findings across multiple cycles or departments. Third, the interview participants were handpicked by the case company rather than randomly selected, meaning their perspectives may not reflect the views of other sectors or functions within the organization. Fourth, the delegated acts for cutting tools under the ESPR have not yet been published, creating inherent regulatory uncertainty. Finally, the interviews were conducted at a single point in time, so longitudinal changes are not captured.

6.6 Final conclusion

Based on the current regulatory framework and the findings of this study, a single carbon footprint value is unlikely to adequately represent the environmental performance of cutting tools. Instead, it will demand a system of verifiable, use phase relevant performance parameters supported by a Digital Product Passport infrastructure (European Parliament and Council, 2024b) and (Battery Pass consortium, 2024a). Companies that act proactively by integrating the four measurable parameters into their product development processes, and by learning from the data architecture of the battery industry and the fair evaluation logic of the steel industry, will turn regulatory compliance into a competitive advantage (ResponsibleSteel and LESS, 2025b).

For Seco Tools, the path forward is not to wait for delegated acts to be published. The interviews consistently showed that the company is already taking important steps, including participation in Digital Product Passport pilot projects, integration of sustainability criteria into gate decisions, and development of application specific Life Cycle Assessment approaches according to the learnings from the interviews. However, significant challenges remain, particularly regarding traceability of regulated substances, supplier data consistency and the fragmentation of environmental data across multiple internal systems according to the Sustainability Compliance Officer and the Sustainability Expert.

Rather than treating the parameters as fixed requirements, they should be used as guidance for what evidence needs to be collected, verified, and connected to future Digital Product Passport requirements. The company should therefore continue developing how these measurement areas can be linked to gate decisions, from early concept development to commercialization Cooper, 2025) and (Seco Tools AB, 2026b). By doing so, Seco Tools can maintain its position as an early mover in the industry and turn regulatory preparedness into a genuine competitive advantage.

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Appendices: Interview questions

Product management team

1. When you're developing new products, which ecodesign criteria, like durability, repairability or recyclability are currently top of mind for you? Have these priorities shifted now that the ESPR is coming into play?

2. Are you using any digital tools, such as CAD integrations or material databases, to check how ready your products are for future DPP requirements? Are there any digital capabilities you wish you had but don't?

3. Let's say you must choose between two materials that are technically similar, one is easier to trace but costs more, while the other is less traceable but cheaper. How do you weigh those trade-offs now, and do you expect that to change in the next few years?

4. Have you had situations where the ESPR's push for higher traceability runs up against your customers' demands for lower prices or specific technical properties? Can you share a concrete example of how you handled that?
5. Of all the product data required for the DPP, which type is hardest for you to generate or verify early in the development process?

R&D Engineer Team *(Research and development)*

1. What are the biggest measurement challenges you see when it comes to quantifying and certifying the percentage of recycled material in a product batch, especially in real time or right next to production?

2. Are there any current R&D projects in your organization that specifically address ESPR related needs, like new traceability methods (for instance, isotope analysis, spectroscopy, or digital twins)?

3. How confident are you in today's sensor technology and data analysis methods for distinguishing and tracing different sources of recycled materials, such as scrap from different suppliers, after they've been melted together?

4. Given the realities of metallurgy, what level of traceability is possible in products made from recycled material with unknown origins? Is there always going to be a degree of uncertainty?

5. What kind of technical breakthroughs would it take to move from batch level traceability to being able to trace every single product individually?

6. Can you walk me through your general product development process? At what points do you consider environmental and sustainability requirements? Are there cases where a project could be stopped if it does not meet your sustainability goals?

7. Develop a product, develop a data, dev an insert, technical center, measure tool life 10-30 min.

8. Without specific delegated acts for cutting tools, companies risk building frameworks that are either too complex or too basic. How do you build flexibility into your product

development process so that R&D doesn't have to start from scratch once the exact requirements are published?

Sustainability Expert Team

1. Looking at the ESPR, whether specific articles, annexes or the expected delegated act for steel. Which parts do you find hardest to interpret right now? How do you deal with that uncertainty within your team?

2. What's your process for checking that the data from product development, R&D, and production really meets the ESPR's requirements for accuracy and completeness? Is there an internal audit or review involved?

3. Which external standards or certifications, like ISO 14021, EN 15804, or Responsible Steel, do you use to help meet ESPR data requirements? Where do you see gaps between these standards and what ESPR is asking for?

4. Compared to other companies in the industry, where do you think Seco Tools is ahead and where is there room for improvement? Are there areas where you have chosen to take a different approach than your peers?

5. If you had to pick one data gap, the one piece of information you cannot reliably produce right now, would be the biggest obstacle to full ESPR compliance?

6. How are Seco Tools preparing, strategically, for the ESPR? Internally, what do you see as your biggest challenge in meeting the requirements for Digital Product Passports (DPP)?

7. We are looking at the battery and steel industries (e.g., the EU Battery Regulation and LESS) as precedents for upcoming regulations. Based on your experience, what practical or data-related pitfalls from these industries should Seco Tools avoid as we prepare for the ESPR?

8. A key part of our thesis is identifying measurable use-phase performance parameters. When you look at metrics like carbon footprint per component, carbon footprint per volume of material removed, or overall tool life – which of these do you see as the most critical, yet most challenging, to accurately measure and report?